True North

MD2 1.538m 163° 9' 47" −1° 23' 2" 701.67 701.43 −16.94

BASE name	BASE height	TARGET name	TARGET height	Azimuth dd	Azimuth mm	Azimuth ss	Elevation dd	Elevation mm	Elevation ss	S	H	V
		1	—	279	12	53"	1	6'	45"	81.17	81.15	1.57
		2	—	284	56	23	0	53	32	80.09	80.07	1.25
		3	—	297	2	38	1	9'	54"	78.09	78.07	1.59
		4	—	302	55	7	0	56	20	77.20	77.19	1.26
		5	—									
		6	—	321	37	35	0	56	57	75.19	75.17	1.25
		7	—	334	36	35	1	13	22	74.24	74.22	1.58
		8	—	340	46	57	0	57	38	73.99	73.98	1.24
		9	—	353	59	27	1	15	33	73.86	73.84	1.62
		10	—	0	10'	37"	0	58'	26"	73.98	73.96	1.26
		11	—	13	14'	37"	1	13'	30"	74.68	74.66	1.59
		12	—	19°	19'	31"	0	57'	45"	75.15	75.14	1.26
		13	—	32°	8'	0"	1°	12'	17"	76.41	76.39	1.61
		14	—									
		CXF	+1"	170°	48'	59"	8°	12'	27"	11.89	11.77	1.70

BASE		TARGET					(feet) distance		
name	height	name	height	azimuth	elevation	H	V	S	note
BR 1	1.350 m	BR2	0	0° 0' 0"	-1° 39' 26"	198.30'			cube
	"	m0	0	72° 23' 8"	0° 11' 48"	227.12	0.80		
				72° 23' 8"	0° 11' 40"	227.12	0.84	227.12	
	"	m1	0	72° 26' 38"	3° 27' 18"	blocked by support	0.77		
	"	m2	0	68° 50' 43"	0° 11' 50"				
				68° 50' 44"	0° 17' 34"				
				68° 50' 46"	0° 17' 25"				
	1.237 m	m3	0	68° 42' 58"	3° 36' 54"	224.08	14.16	224.53	
	"	m4	0	62° 29' 29"	0° 16' 58"				
	"	m5		62° 34' 55"	3° 30' 4"	231.35	14.15	231.78	
	"	m0		72° 23' 4"	0° 17' 10"	227.11	1.13	227.11	
	"	m2		68° 50' 24"	0° 17' 21"	222.96	1.13	222.96	
	"			62° 29' 17"	0° 17' 6"	230.24	1.15	230.24	
	"	BR2		0° 0' 0"	-1° 33' 11"	198.20	5.37	198.27	
BR2	1.441 m	BR1		0° 0' 0"	-59' 19"	198.21	3.42	198.284	

				azi	elev	H	V	S
BR2	1.441	m5	0	297°30'8"	0°26'5"	—		
"	"	m6	0	297°27'38"	3°39'11"	231.38	14.77	231.85
	"			291°15'27"	3°46'33"	223.97	14.78	224.45
		blank						
	"			287°36'26"	0°26'43"	227.08	1.77	227.08
	"			291°8'38"	0°27'23"	—		

	BASE Station		Target Station		Target Azimuth				Target Elevation					Target Distance (feet)			Notes
	name	height (m)	name	height (m)	dd	mm	ss	ddd.ddddddd	sgn	dd	mm	ss	ddd.ddddddd	S	H	V	
0	MD1	1.535	CXF	0.025	67	39	7	67.651944	1	8	12	27	8.207500	11.89	11.77	1.70	
1	BR1	1.324	ELS	1.896	320	53	41		+	3	2	58		189.08	188.81	10.06	cube
2	"	"	BR2	0	0	0	0	—	−	1	38	0		198.28	198.20	5.65	
3	"																
4	BR2	1.304	BR1	0	0	0	0		−	0	51	3		198.23	198.20	2.95	
5	BR2	"	ELS	1.896	0	0	0		+	5	2	0					
6	"	"	"	"	"	"	"		+	5	1	54		130.37	129.87	11.43	cube
7	BR1	1.392	BR2	0	0	0	0		−	1	42	4		198.29	198.20	5.89	
8			M2	0	72	23	0		+	0	9	31		227.12	227.11	0.63	
9			M3	−7 1/8"	72	20	5		+	3	16	2		228.67	228.30	13.02	
10			M1	0	68	42	48		+	3	29	5		224.49	224.06	13.64	
11			M0	0	68	50	26		+	0	9	34					
12			M0	−7 1/8"	68	50	32		+	0	0	36		222.86	222.86	0.04	
13			M4	0	62	29	3		+	0	9	26					
14			M5	0	62	34	44		+	3	22	28		231.75	231.36	13.64	
15			M4	−7 1/8"	62	29	4		+	0	0	34		230.14	230.14	0.04	
16			M4	+7"	62	29	11		+	0	18	20		230.30	230.29	1.23	
17			M5	0	62	34	52		+	3	22	31		231.76	231.36	13.64	
18	LR2	1.424	LR1	0	0	0	0		−	1	50	57		194.21	194.11	6.27	
19			m3	0	73	56	46		+	7	28	21		223.38	221.48	29.04	
20			m1	0	70	11	58		+	7	37	9					
21			m5	0	63	52	20		+	7	22	20					
22			m2	+7"	73	59	27		+	4	19	27		221.02	220.39	16.67	
23			m0	+7"	70	20	24		+	4	24	25		216.87	216.22	16.66	
24			m1	+7"	70	12	22		+	7	46	21		219.17	217.15	27.64	
25			m1	−7 1/8"	70	11	41		+	7	27	32		219.06	217.22	28.44	
26			m4	+7"	63	46	44		+	4	15	58		224.25	223.63	16.68	
27			m5	−7 1/8"	63	52	42		+	7	13	12		226.49	224.69	28.46	
28			m5	+7"	63	52	23		+	7	31	20		226.60	224.65	29.66	
29	LR1	1.470	LR2	0	0	0	0		−	0	56	14		194.13	194.09	−3.18	
30			m9	0	288	50	36		+	7	41	11		228.66	226.60	30.58	
31			m11	0	292	27	35		+	7	49	29					

	BASE Station		Target Station		Target Azimuth				Target Elevation					Target Distance (feet)			Notes
	name	height (m)	name	height (m)	dd	mm	ss	ddd.ddddddd	sgn	dd	mm	ss	ddd.ddddddd	S	H	V	
0	MD1	1.535	CXF	0.025	67	39	7	67.651944	1	8	12	27	8.207500	11.89	11.77	1.70	
1	LR1	1.470	m8	+7 1/8	288	47	17		+	4	36	47		226.20	225.47	18.19	
2			m10	+7 1/8	292	20	23		+	4	41	53		221.98	221.24	18.18	
3			m11	+7 1/8	292	27	23		+	7	58	44		224.34	222.16	31.14	
4			m11	-7 1/8	292	27	23		+	7	40	22		224.23	222.22	29.94	
5			m7	0	298	38	58		+	7	34	55		231.51	229.49	30.54	
6			m6	+7 1/8	298	44	28		+	4	33	08		229.14	228.14	18.19	
7	BR2	1.390	m8	0	287	36	28		+	0	29	36		227.09	227.07	1.95	
8			m10	0	291	8	29		+	0	30	11					
9			m11	0	291	15	15		+	3	49	20		224.46	223.97	—	
10			m9														blocked
11			m10	+7 1/8	291	9	7		+	0	39	22		222.99	222.97	2.55	
12			m10	-7 1/8	291	08	10		+	0	21	2		222.78	222.78	1.36	
13			m10	0	291	8	33		+	0	30	20					
14			m9	-7 1/8	287	39	40		+	3	36	15		228.67	228.22	14.37	
15			m9	+7 1/8	287	39	40		+	3	54	25		228.70	228.17	15.58	
16			m6	-7 1/8	297	29	53		+	0	20	1		230.11	230.11	1.34	
17			m6	+7 1/8	297	29	53		+	0	38	6		230.29	230.29	2.55	
18			m7	0	297	25	51		+	3	41	57					
19			m7	-7 1/8	297	25	51		+	3	32	47		231.80	231.36	14.34	
20			m7	+7 1/8	297	25	51		+	3	50	50		231.83	231.31	15.55	
21			BR1	0	0	0	16		−	0	56	9		198.24	198.21	-3.24	
22	MD1	1.527	MD2	1.675	0	0	0		−	1	14	23		701.59	701.42	-15.17	
23			m7	0	231	26	43		+	1	14	51		74.24	74.23	1.62	
24			m8	0	237	37	7		+	0	59	10		73.79	73.98	1.27	
25			m9	0	250	49	44		+	1	17	19		73.87	73.84	1.66	
26			m10	0	257	0	58		+	1	0	17		73.97	73.96	1.30	
27			m11	0	270	4	55		+	1	15	12		74.68	74.66	1.63	
28			m12	0	276	9	43		+	0	59	27		75.15	75.14	1.30	
29			m13	0	288	58	34		+	1	13	49		76.41	76.39	1.64	
30			m14	0	294	54	36		+	0	58	43		77.22	77.21	1.32	partially blocked
31			m6	0	218	28	1		+	0	58	33		75.19	75.18	1.28	

	BASE Station		Target Station		Target Azimuth				Target Elevation					Target Distance (feet)			Notes
	name	height (m)	name	height (m)	dd	mm	ss	ddd.ddddddd	sgn	dd	mm	ss	ddd.ddddddd	S	H	V	
0	MD1	1.535	CXF	0.025	67	39	7	67.651944	1	8	12	27	8.207500	11.89	11.77	1.70	
1	MD1	1.527	m5	0	212	24	54		+	1	12	54		75.78	75.71	1.61	partial block
2			m4	0	199	45	15		+	0	57	50		77.19	77.19	1.30	
3			m3	0	193	52	26		+	1	11	37		78.08	78.06	1.63	
4			m2	0	181	40	27		+	0	54	48		80.08	80.07	1.30	
5			m1	0	176	2	34		+	1	8	19		81.09	81.07	1.64	
6			CXF	+1¼"?	67	39	28		+	8	22	58		11.90	11.77	1.74	
7																	
8																	
9																	
10																	
11																	
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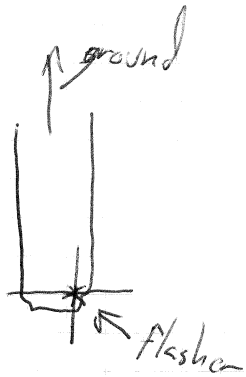
3-22-2011 Surveying @ BRFD SBT



19:21 A2 Gyro

Base BRFD#2 Marker

Target IPP smokestack flasher

~~1595~~	3510	0310
~~2.48~~	3.22	3.34
~~1597.48~~		
	17°37'7"	17°37'31.3"

$\boxed{\angle 7 = 17.622 = 17°37'19.2'}$

Set 17°37'19" to smoke stack! $\boxed{3510 \quad 3.22}$ $\boxed{0310 \quad 3.34}$ repeat,

Polaris 21:09:55

359°9'15" 39°23'38"

Stellarium 359°8'45" 39°22'42"

21:21:20 359°8'32" 39°21'29"

Stellarium 359°8'6" 39°20'42"

Smoke stack 17°37'19"! check.

BRFP#1 206° 22' 31"

3-23-2011 JBT

1. Set up GPS @ LRFD #1.

BASE	Target	azi	elev	dist
BRFD #1	BRFD #2	0°0'0"	1°39'26"	H: 198.30'
H=1.350 m				H: 198.30'
				H: 198.31'
				horiz dist

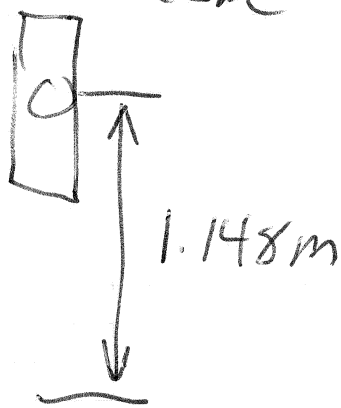
3-24-2011 SBT

MD1 azimuth - gyro
ref: peak. 0640
3.52

3840
3.48

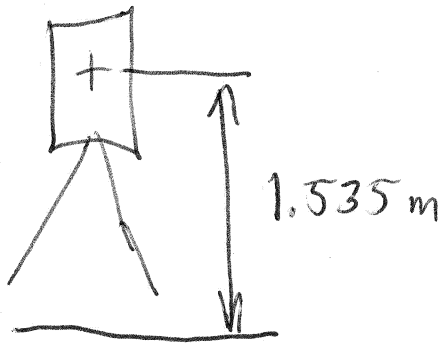
216° 11' 48" ✓

Target cube

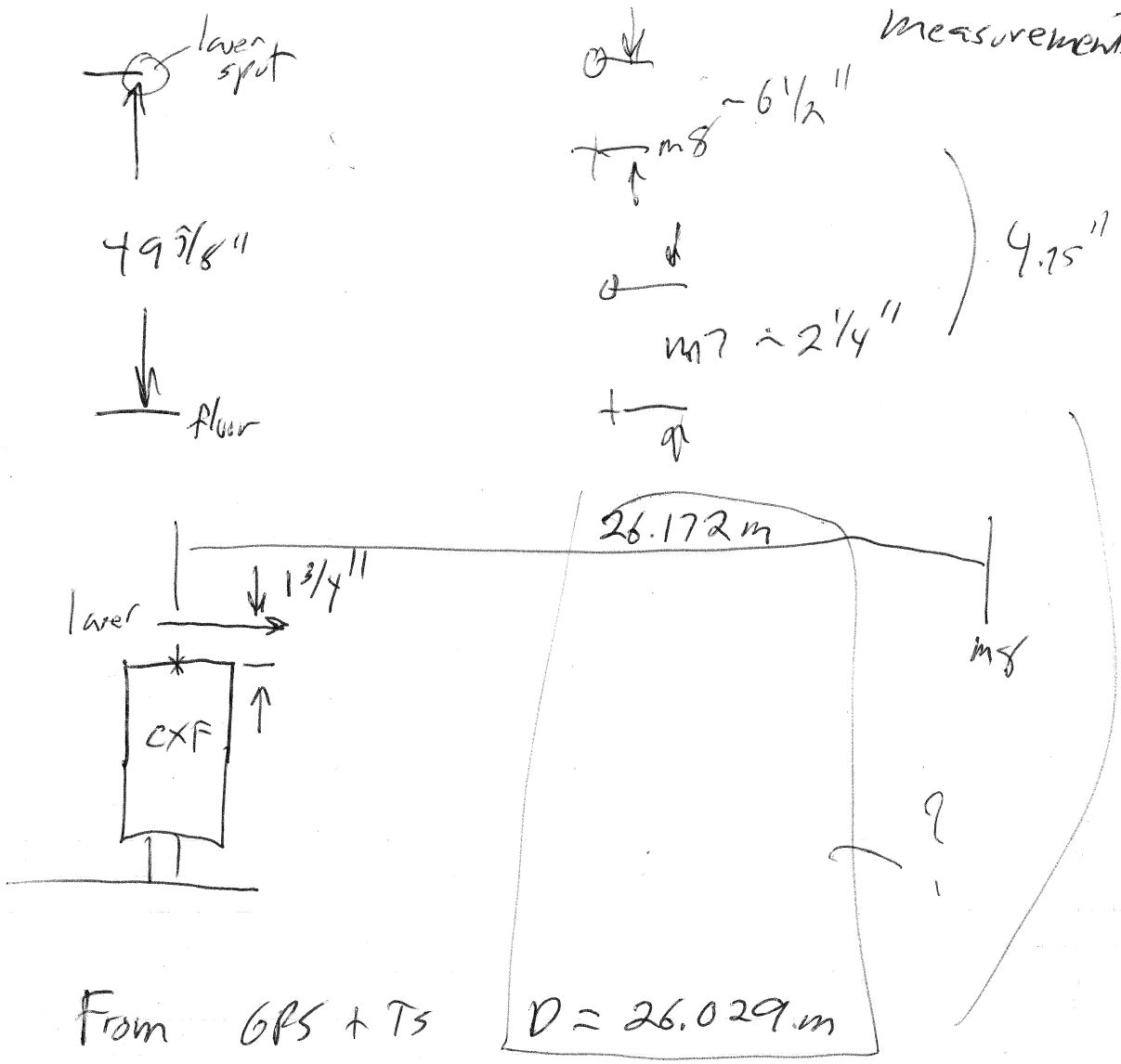


MD2 Azimuth
103° 9' 57"

S	H	V
701.64	701.42 43	-16.95



3-29-2011 MD SBT Cross check measurements.



From GPS + TS $\boxed{D = 26.029\,m}$

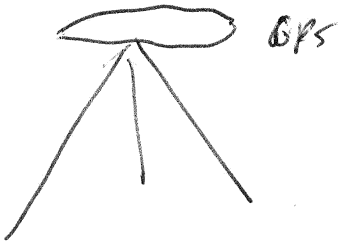
$D_e = -8.0052$

$D_n = 24.7669$

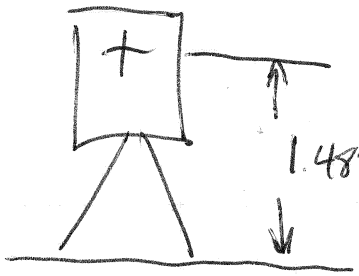
$D_u = -0.1141$ 4.5"

3-30-2011 CLF2 S.B.T.

CLF2

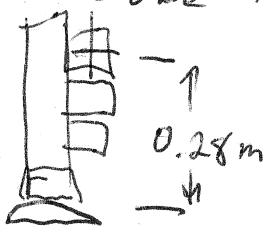


CLF1 set3 Total station



1.489m, 1.481m, 1.480, 1.488 (concrete)

cube target



S 591.97'

H 591.95'

V 4.11'

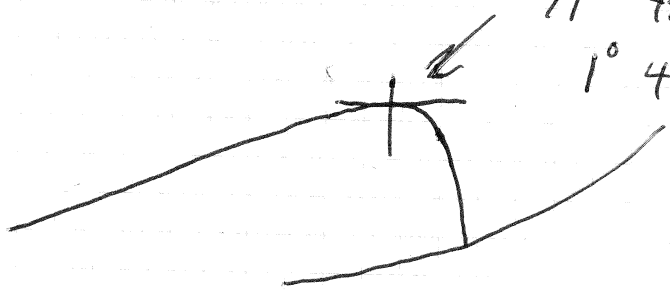
CLF H 55.15' 55.16' 55.15'

Azi = (215° 51' 7" + 216° 38' 29")/2

~ 216° 15' 40"

Notch Peak

71° 48' 44" azi wrt CLF2
1° 43' 38" elev



IPP light 233° 3' 13"

CLF 2 1.411 to bottom of notch

3-30-2011 SBT.

LR2 1.425 m to bottom of notch

LF2

3-31-2011 SBT.

BR 2 OBS

9	11	5	7	1	3
8	10	4	6	0	2